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Handling Missing Data for Downstream Tasks

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Abstract

This the abstract in English.

Zusammenfassung

Diese ist die Zusammenfassung auf Deutsch.

Theses

These are the theses.

Table of contents

Acknowledgements	iii
Abstract	iv
Zusammenfassung	v
Theses	vi
List of Figures	ix
List of Tables	x
List of Acronyms	xi
Summary and Outline of Thesis	1
Summary	1
Outline	1
1 Challenges with Missing Data	4
1.1 Notation	4
1.2 Problem Formulation	5
1.3 Types of missingness	6
1.3.1 MCAR	6
1.3.2 MAR	6
1.3.3 MNAR	6
2 State-of-the Art in Imputation	7

3	Error Guarantees for Nearest-Neighbor Imputation	8
4	Multivariate Functional Linear Discriminant Analysis for Partially-Observed Time Series	9
5	Fast Semi-Supervised Node Embedding via Structural and Label-Aware Optimization	10
6	Conclusion	11
	References	12
	Appendices	13
A	Technical Proofs	13

List of Figures

List of Tables

List of Acronyms

F3I Fast Iterative Improvement for Imputation

FUSE Fast Semi-Supervised Node Embedding via Structural and Label-Aware Optimization

MAR Missing At Random

MCAR Missing Completely at Random

MNAR Missing Not at Random

MUDRA Multivariate Functional Linear Discriminant Analysis

kNN K-Nearest Neighbours

Summary and Outline of Thesis

Summary

Outline

- Chapter 1
 - Notation used
 - Sources of missingness: improper data collection, malfunctioning equipment, privacy issues
 - Types of missingness: [Missing Completely at Random \(MCAR\)](#), [Missing At Random \(MAR\)](#), [Missing Not at Random \(MNAR\)](#)
 - * Models of these missingness mechanisms: Bernoulli, logistic, Gaussian self-masking, logistic self-masking
 - Imputation: advantages & disadvantages
- Chapter 2
 - Constant imputation: constant value, measures of central tendency
 - Multivariate Imputation: Amelia, mi, MICE, missForest
 - Generative methods: GAN-based models, diffusion-based models, transformer-based models
 - Bayesian methods: MIWAE, not-MIWAE.
 - * Designed to optimize for downstream tasks
 - * Aligns poorly with noisy ground truth missing values
 - * Morvan and Varoquaux [\[2025\]](#) shows that higher quality imputation does not necessarily improve downstream task performance
 - Nearest neighbour imputation: looking at closer observed values gives a better idea of the missing value
 - Ensemble methods: HyperImpute

- Can nearest neighbour imputation be modified to optimize for downstream tasks like MIWAE, while maintaining alignment with the data?
- Chapter 3
 - Reintroduce nearest neighbour imputation, and state objective to modify it for downstream task
 - Argue why it is better than Bayesian methods in maintaining alignment with data, show high probability concentration bounds for imputed values.
 - Introduce [Fast Iterative Improvement for Imputation \(F3I\)](#) and the novel objective function. Argue why a differentiable function makes it easier to modify and optimise for downstream task.
 - Present PCGrad[F3I](#).
 - Conclusion: reiterate that better imputation does not necessarily mean better downstream task performance. Argue for the need of methods to perform downstream tasks without imputation.
- Chapter 4
 - Restate problem with imputation according to Morvan and Varoquaux [\[2025\]](#)
 - Introduce the multivariate time-series classification problem.
 - Present [Multivariate Functional Linear Discriminant Analysis \(MUDRA\)](#) [\[Bordoloi et al. 2025\]](#) to perform dimension reduction and classification without imputation.
 - Show how classification performance is improved over LOCF imputation
 - Show how imputed values closely align with ground truth
 - Conclusion: We talked about missing values on tables, and missing values on multivariate time series. Now we introduce missing values on graph clustering, a more complicated data structure.
- Chapter 5
 - State the semi-supervised learning problem on graphclustering: learning when labels are missing, which means modularity cannot be directly calculated
 - Show how to use the graph structure to generate representations even with missing modularity, by imputing labels so as to increase the modularity of the remaining graph
 - Show how the labels can then be reconstructed time-efficiently.
- Chapter 6
 - Conclude that missing values are best dealt with without imputation
 - The more complicated the underlying data structure, the more information to be gleaned to bypass imputation.

-
- MArginal gains in imputation are not worth it, instead imputation should be moderated by downstream task performance in order to get the best results.

Chapter 1

Challenges with Missing Data

In most statistical and machine learning literature, datasets are assumed to be complete, regular, and unfragmented. In short, most models assume complete data for processing. However, in the real domain, a lot of datasets are incomplete, irregular, and fragmented. This is particularly evident in the medical and biomedical domain, where collection of datasets are dependent on the availability of patients and the robustness of wet lab experiments, which are not always 100% reliable.

In this chapter, I will introduce the different types of missingness that has been proposed in the literature, as well as propose a new classification of missingness based on a different perspective. I will also discuss how the semi-supervised setting can be viewed as a special case of missing data, and how it can be incorporated into the overall missingness framework.

1.1 Notation

Throughout this thesis, I will use the standardised notation encouraged by most machine learning conference. Italicized letters will indicate deterministic quantities. Normal small letters indicate numbers (x, y, t), bold small letters indicate vectors ($\mathbf{x}, \mathbf{y}, \mathbf{t}$), bold capital letters indicate matrices ($\mathbf{X}, \mathbf{Y}, \mathbf{T}$), capital letters in sans-serif fonts indicate tensors (X, Y, T) and blackboard bold capital letters indicate sets ($\mathbb{X}, \mathbb{Y}, \mathbb{T}$). The non-italicized versions of these indicate random quantities; random variables are x, y, t , random vectors are $\mathbf{x}, \mathbf{y}, \mathbf{t}$ and random matrices are $\mathbf{X}, \mathbf{Y}, \mathbf{T}$. Elements of higher order arrays like vectors, matrices, and higher order tensors are denoted by regular versions of the bold letters with subscripts: $x_1, X_{11}, X_{111}, x_1, X_{11}$, etc. For a matrix $\mathbf{X}$, $\mathbf{X}_{i,:}$ and $\mathbf{X}_{:,i}$ will denote the i^{th} row and column respectively.

I will denote the vectorization of a matrix $\mathbf{X}$, which means stacking the columns of the matrix sequentially, as

$$\vec{(\mathbf{X})} = [\mathbf{X}_{:,1}^\top \quad \mathbf{X}_{:,2}^\top \quad \cdots \quad \mathbf{X}_{:,n}^\top]^\top$$

The determinant of a matrix $\mathbf{A}$ will be denoted by $|\mathbf{A}|$, and Kronecker product of two matrices $\mathbf{A}_{m \times n}$ and $\mathbf{B}_{p \times q}$ is an $mp \times nq$ matrix defined as

$$\mathbf{A} \otimes \mathbf{B} \triangleq \begin{bmatrix} A_{11}\mathbf{B} & A_{12}\mathbf{B} & \cdots & A_{1n}\mathbf{B} \\ A_{21}\mathbf{B} & A_{22}\mathbf{B} & \cdots & A_{2n}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}\mathbf{B} & A_{m2}\mathbf{B} & \cdots & A_{mn}\mathbf{B} \end{bmatrix}$$

The generalized psedo-inverse [Rao and Mitra 1971, p. 20] of a matrix $\mathbf{A}$ will be denoted by $\mathbf{A}^-$, which simply indicates that $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$. It should be clear that the choice of $\mathbf{A}^-$ is not unique for a singular matrix $\mathbf{A}$, *i.e.* if $|\mathbf{A}| = 0$. In cases where a unique choice is necessary, I will use the Moore-Penrose pseudo-inverse [Penrose 1955] $\mathbf{A}^+$, which is defined by the following properties:

$$\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A} \quad \mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+ \quad (\mathbf{A}\mathbf{A}^+)^* = \mathbf{A}\mathbf{A}^+ \quad (\mathbf{A}^+\mathbf{A})^* = \mathbf{A}^+\mathbf{A}$$

I will denote the probability of an event E as $\mathbb{P}(E)$, expectation of a random variable x as $\mathbb{E}[x]$, its variance as $\text{Var}(x)$, and its covariance with another random variable y as $\text{Cov}(x, y)$. I will denote the variance-covariance matrix of a random vector $\mathbf{x}$ as $\text{Var}(\mathbf{x})$. For a matrix Σ , $\|\mathbf{x}\|_\Sigma$ will denote the Mahalanobis norm [Mahalanobis 2018] of $\mathbf{x}$ with covariance Σ , defined as $\|\mathbf{x}\|_\Sigma \triangleq \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$.

I will also denote the KL divergence of 2 distributions P and Q over the support $\mathbb{X}$ as

$$D_{\text{KL}}(P||Q) \triangleq \int_{x \in \mathbb{X}} P(x) \log \frac{P(x)}{Q(x)} dx$$

The Cartesian product of two sets $\mathbb{A}$ and $\mathbb{B}$ will be denoted by $\mathbb{A} \times \mathbb{B}$.

1.2 Problem Formulation

I will adopt notation similar to Jarrett et al. [2022] for the problem formulation and denote missing values as NaN. Let $\mathbf{x} \in \mathbb{X} = (\mathbb{X}_1 \times \dots \times \mathbb{X}_D)$ be a D -dimensional random vector. The

entries of this random vector can be continuous ($\mathbb{X}_d = \mathbb{R}$) or categorical ($\mathbb{X}_d = \{1, \dots, K_d\}$) random variables, for $d \in \{1, \dots, D\}$. We do not have complete observational access to $\mathbf{x}$; instead our observation is mediated by a random masking vector $\mathbf{m} \in \{0, 1\}^D$. If $m_d = 1$, then x_d is observed. Assume that $\mathbf{m}$ follows a distribution p_{mask} , and the data $\mathbf{x}$ follows a distribution p_{data} .

1.3 Types of missingness

Classical literature mainly distinguishes three main missingness mechanisms depending on the relationship between the probability of missingness with the observed and unobserved data [Rubin 1976]. These can be distinguished based on the form and properties of p_{mask} .

1.3.1 MCAR

MCAR is a missingness mechanism where the probability of missingness is independent of the observed and unobserved data. In other words, the missingness is completely random and does not depend on any of the observed or unobserved data. This is equivalent to the statement that $\mathbf{x} \perp\!\!\!\perp \mathbf{m}$. This also indicates that each component of $\mathbf{m}$, m_i , is an independent Bernoulli random variable with probability p_i independent of $\mathbf{x}$.

1.3.2 MAR

The missingness mechanism **MAR** indicates that the probability of missingness depends only on the observed values. In more technicality, let $\mathbf{x}$ and $\mathbf{m}$ be partitioned as $\{\mathbf{x}_1, \mathbf{x}_2\}$ and $\{\mathbf{m}_1, \mathbf{m}_2\}$ respectively, such that $\mathbf{x}_1$ is always observed, that is, $\mathbf{m}_1 = \mathbf{1}$. Then, $\mathbf{x}$ is said to be **MAR** if p_{mask} depends only on $\mathbf{x}_1$.

1.3.3 MNAR

If data is missing by the **MNAR** mechanism

Chapter 2

State-of-the Art in Imputation

Chapter 3

Error Guarantees for Nearest-Neighbor Imputation

Chapter 4

Multivariate Functional Linear Discriminant Analysis for Partially-Observed Time Series

Chapter 5

Fast Semi-Supervised Node Embedding via Structural and Label-Aware Optimization

Chapter 6

Conclusion

This is the conclusion of the thesis.

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Appendix A

Technical Proofs

This is the appendix of the thesis.